LLMs predict metaphor norms based on word co-occurrence but struggle with relational and sensory dimensions

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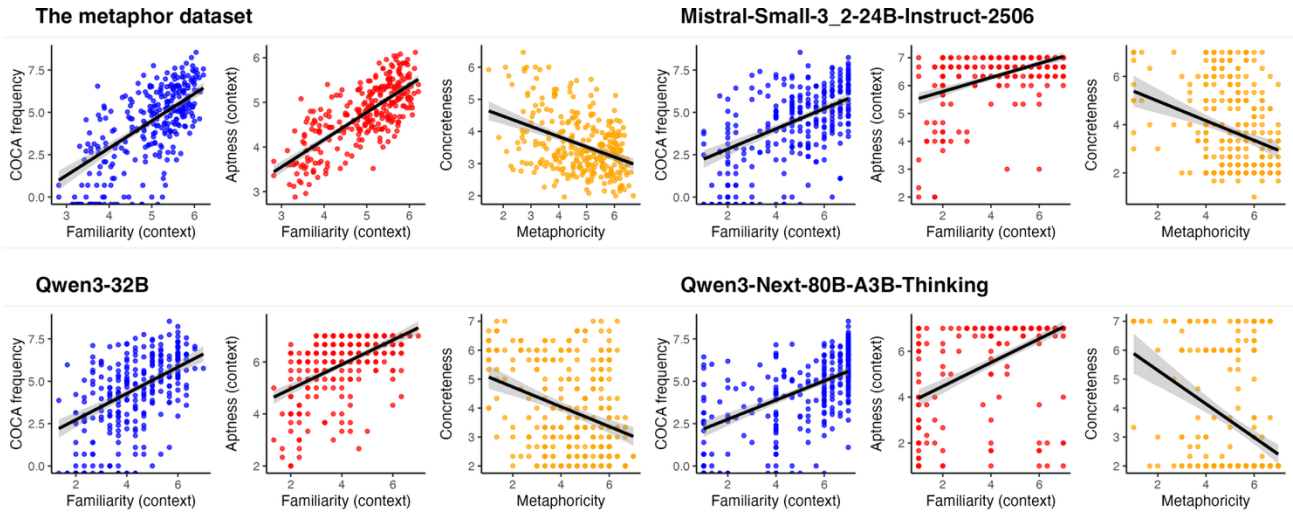
Introduction. Research in psycholinguistics relies on normed linguistic stimuli, which often include lexical (e.g., frequency, length) and semantic (e.g., familiarity, aptness) properties known to affect comprehension. Such norms are essential for accounting for factors that may influence participants' responses. However, building normed datasets at scale is time-consuming and costly, often leading researchers to create *ad hoc* datasets with smaller numbers of participants, which can result in inconsistencies and limit comparisons across studies. **Current study.** To address these limitations, we investigated the use of large language models (LLMs) to generate norms for psycholinguistic research. We prompted eight current LLMs [1-8] to rate 300 two-word English metaphors (e.g., *sharp mind*) on familiarity, aptness, concreteness, metaphoricity, and constituency from a recent metaphor dataset [9]. We extended previous work by manipulating stimuli presentation (in context vs. in isolation) to examine whether LLMs' ratings are affected by context similar to human norms [9]; type of output generated (categorical vs. graded) to assess whether LLMs perform better when producing numerical ratings or assigning categories; and including metaphor-specific dimensions to evaluate how models generate norms for dimensions based on frequency of use (e.g., familiarity) compared to those requiring reasoning about the topic-vehicle mapping (e.g., aptness), sensory or physical experience (e.g., concreteness), contextual information (e.g., metaphoricity), or analysis of phrase structure (e.g., constituency). We first assessed the reliability and validity of LLM-generated norms by measuring internal consistency across three random seeds and alignment with human norms for the full set of 300 metaphors and for a reduced set of 24 metaphors, used in an online study of metaphor comprehension [10]. We then conducted substitution analyses to examine whether LLM-generated norms replicated the relationship between dimensions reported in the metaphor dataset [9] and effects observed in the online study [10]. Finally, we examined the generalizability of our findings by applying the prompts used to generate familiarity ratings to a separate set of metaphors used in an offline study [11], which normed their materials using ratings from a different metaphor dataset [12]. **Results.** Table 1 shows that reliability was consistently high across all models and dimensions, except for constituency and vehicle position—both assessing whether metaphorical content is in the modifier, head, or both words of the expression, measured numerically and categorically, respectively. Most models showed moderate to strong correlations with human ratings. Familiarity and metaphoricity showed the strongest alignment, while aptness, concreteness, and constituency, the weakest. The inclusion of context did not noticeably affect familiarity or aptness ratings. Interestingly, for half of the models, constituency showed good reliability and moderate to strong alignment with human ratings, whereas vehicle position consistently showed poor reliability and alignment—despite both measuring the same construct. Overall, norms from all LLMs replicated the relationships reported in the metaphor dataset, although model responses exhibited more extreme values (see Fig. 1). Replication of effects from the online study was more variable, with most models replicating one or two of the three findings. For the second dataset, six models were selected based on their reliability and validity with the new norms [12], and of these, only Qwen3-32B and Qwen3-Next-80B-A3B-Thinking replicated the familiarity effects [11]. **Discussion.** Altogether, LLMs reliably predict dimensions based on word co-occurrence (i.e., familiarity, metaphoricity) but are less consistent for dimensions that require relational reasoning about metaphor components (i.e., aptness, constituency) or sensory experience (i.e., concreteness). LLM-generated norms should therefore be used only with careful validation, as even highly reliable estimates may fail to replicate experimental effects.

Table 1. Reliability was measured using intraclass correlation coefficients (ICC) for ratings and Krippendorff's alpha (K's α) for categorical responses. For models, reliability was computed across three random seeds; for humans, values were obtained from the metaphor dataset [9] across multiple participants. Alignment with human norms was measured using Spearman's rho (ρ) for ratings and Cohen's kappa (C's k) for categorical responses. Values below commonly accepted thresholds (ICC < 0.50, K's α < 0.67, C's $k \leq 0.40$) are shown in gray, and asterisks denote statistical significance ($p < .01$). Dimensions assessed include familiarity in context (FAMc), familiarity in isolation (FAMi), aptness in context (APTc), aptness in isolation (APTi), concreteness (CON), metaphoricity (MET), constituency (CONS), and vehicle position (Vehicle).

	FAMc		**FAMi**		**APTc**		**APTi**		**CON**		**MET**		**CONS**		**Vehicle**	
Model	ICC	ρ	ICC	ρ	ICC	ρ	ICC	ρ .	ICC	ρ	ICC	ρ	ICC	ρ	K's α	C's k
Met. dataset	0.88	—	0.88	—	0.78	—	0.70	—	0.87	—	0.94	—	0.93	—	—	—
Apriel	0.88	0.50*	0.88	0.47*	0.80	0.18*	0.95	0.27*	0.68	0.30*	0.87	0.62*	0.88	0.67*	0.30	0.00
Mistral	0.96	0.64*	0.96	0.66*	0.92	0.27*	0.96	0.31*	0.93	0.43*	0.90	0.65*	0.35	0.09	0.46	-0.01
Qwen3	0.88	0.66*	0.90	0.64*	0.86	0.27*	0.92	0.31*	0.84	0.49*	0.93	0.67*	0.48	0.40*	0.34	0.00
Apertus	0.86	0.62*	0.84	0.56*	0.60	0.23*	0.63	0.25*	0.72	0.42*	0.60	0.45*	0.43	0.03	0.16	-0.01
Llama3	0.94	0.66*	0.94	0.65*	0.86	0.29*	0.91	0.34*	0.89	0.37*	0.93	0.68*	0.39	0.28*	0.37	0.02
Qwen3-Next	0.98	0.75*	0.97	0.65*	0.97	0.38*	0.95	0.36*	0.97	0.50*	0.94	0.61*	0.98	0.76*	0.79	-0.02
gpt-oss	0.93	0.57*	0.96	0.57*	0.91	0.27*	0.95	0.27*	0.83	0.44*	0.91	0.65*	0.79	0.62*	0.29	0.00
DeepSeek	0.97	0.74*	0.96	0.71*	0.85	0.25*	0.91	0.34*	0.84	0.39*	0.93	0.72*	0.67	0.64*	0.43	0.00

Figure 1. Correlation plots among dimensions for the metaphor dataset and best-performing LLMs.



References. [1] Radhakrishna et al. (2025); [2] Mistral AI Team (2025); [3] Qwen Team (2025b); [4] Hernández-Cano et al. (2025); [5] Grattafiori et al. (2024); [6] Qwen Team (2025a); [7] OpenAI Team (2025); [8] DeepSeek-AI et al. (2024); [9] Pissani & de Almeida (2026) *Behavior Research Methods*; [10] Pissani, Genovesi & de Almeida (under review); [11] Stamenković, Ichien & Holyoak (2019) *Journal of Memory and Language*; [12] Cardillo, Schmidt, Kranjec & Chatterjee (2010) *Behavior Research Methods.*